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KOLLAM DISTRICT

SSLC

CHEMISTRY

MODEL QUESTIONS (D+ Grade)

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UNIT - 1

Nomenclature of Organic Compounds and Isomerism

(1 Mark Questions)

1. Structure of an organic compound is given

   CH_3-CH_2-C $\equiv$ CH

   This compound belongs to which category among the following.
   (alkane, alkene, alkyne)
2. Find the relation and fill up suitably ?

   -OH : alkanol

   -CHO : …….

   (A) Alkanoic Acid

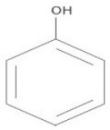
   (B) Alkanone

   (C) Alkanal

   (D) Alkoxy Alkane
3. A homologous series is given

   C_2H_4, C_3H_6, __A__ , C_5H_{10} __B__

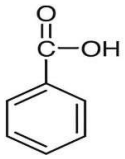
   Find A & B
4. Write the name of the given organic compound



5. Which is the functional isomer of alcohol.

(2 Mark Questions)

1. Structure of an organic compound is given



   (a) This compound belongs to which category of the following
   (Aliphatic, Aromatic, Alicyclic)

   (b) Write the name of this compound
2. The structure of an organic compound is given

   CH_3-CH-CH_2-CH_3

   CH_2-CH_3

   (a) How many carbon atoms are there in the main chain ?

   (b) Write the name of the branch
3. Classify the following compounds into alkanes, alkenes and alkynes

   C_2H_6, C_5H_{10}, C_3H_4, C_6H_{12}

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4. Analyse the given structure

CH_3-CH_2-CH_2-OH

(a) Write the structure of the position isomer of this compound.

(b) Write the structure of the functional isomer of this compound

(3 mark Questions)

1. Analyse the given structure

CH_3-CH_2-CH(-CH_2-CH_3)-CH_2-CH_3

(a) How many Carbon atoms are in the main chain ?

(b) Name the branch

(c) Write the IUPAC name

2. Structure of two organic compound is given

(i) CH_3-CH_2-CH_2-CH_2-CHO

(ii) CH_3-CH_2-CO-CH_2-CH_3

(a) Write the molecular formula of these compounds

(b) Which isomerism is exhibited by these compounds?

(c) Write the IUPAC name of the first compound.

Answers

(1 Mark Questions)

1. Alkyne
2. Alkanal
3. A-C_4H_8
   B-C_6H_{12}
4. Phenol
5. Ether

(2 mark Questions)

1. (a) Aromatic
   (b)Benzoic acid
2. (a) 5
   (c) Methyl
3. Alkane - C_2H_6
   Alkene - C_5H_{10}, C_6H_{12}
   Alkyne - C_3H_4
4. CH_3-CH(-OH)-CH_3
   (c) CH_3-CH_2-O-CH_3

(3 mark Questions)

1. (a) 5
   (b) Ethyl
   (c) 3-Ethyl Pentane
2. (a) C_5 H_{10} O
   (b) Functional Isomerism
   (c) Pentanal

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UNIT – 2

CHEMICAL REACTION OF ORGANIC COMPOUNDS

(1 Mark Questions)

3. Which is the monomer of polyvinyl chloride ?
4. Name the polymer used to coat the inner surface of non-stick cooking vessels.
5. Write an example for condensation polymer?
6. Which compound is known as wood sprit?
7. What is power alcohol?
8. 5-8% ethanoic acid is known as ……
9. Aspirin belongs to which category of medicines?
10. Which ester has the smell of orange?
11. Which are the products formed by the combustion of hydrocarbons?
12. Which polymer is obtained by the condensation polymerisation of phenol and formaldehyde?
13. $CH_2=CH_2+H_2 \longrightarrow CH_3\text{-}CH_3$
    This chemical reaction belongs to which category among the following ?
    (Substitution reaction, Addition reaction, Combustion, Polymerisation)
14. Which is the monomer of natural rubber

Answers

1. Vinyl chloride
2. Teflon
3. Nylon 66/Bakelite/ Polyester
4. Methanol
5. Mixture of absolute alcohol and petrol
6. Vinegar
7. Analgesics
8. Octyl ethanoate
9. CO_2, H_2O
10. Bakelite
11. Addition reaction
12. Isoprene

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UNIT - 3

PERIODIC TABLE AND ELECTRON CONFIGURATION

(1 Mark Questions)

1. Among the following which subshell has greater energy.
   [(a) 3P, (b) 3d, (c) 4S, (d) 2P]
2. n=1 represents which shell ?
3. How many orbitals are for P subshell ?
4. What is the maximum number of electrons that can accommodate in L shell ?
5. What happens to the ionisation enthalpy on moving down the group?
6. Write the oxidation numbers of Fe in $FeCl_2$?
7. At which subshell electron filling takes place in f block elements ?
8. Which element is used to make strong magnets ?
9. Which are the f block elements used as catalyst in petroleum industry ?
10. Lanthanoids belong to which period of the periodic table ?

(2 mark Questions)

1. Subshell electron configuration of an element 'X' is given
   $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^5$
   (a) What is the atomic number ?
   (b) X belongs to which group ?
2. Subshell electron configuration of Cu is given
   (i) $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3d^9\ 4S^2$
   (ii) $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^6\ 3d^{10}\ 4S^1$
   (a) Which is the correct subshell electron configuration?
   (b) Write the reason
3. Write any two characteristics of d block elements?
4. (a) n=2, l=1 denotes which subshell ?
   (c) Write the maximum number of electrons that can accommodate in this subshell?
5. Outer electron configuration of an element is $3d^64S^2$
   (a) Write the complete subshell electron configuration.
   (b) At which block this element belongs ?
6. Find the n & l values of the following subshells ?
   (a) 2S (b) 3d
7. Some subshells are given
   3d, 4d, 3f, 2d, 2p
   (a) Among these which subshells are not possible ?
   (b) Write the reason

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Answers

(1 mark Questions)

1. 3d
2. K shell
3. Three
4. Eight
5. Decreases
6. +2
7. Anti penultimate shell
8. Neodemium
9. Cerium /Lanthanam
10. 6th

(2 mark Questions)

1. (a) 17
   (b)17
2. (a) 11
   (b) Half filled or completely filled d subshells have stability.
3. Variable oxidation state, coloured compounds; All are metals, same properties in period and groups.
4. (a) 2P
   (b) 6
5. (a) $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^6\ 3d^6\ 4S^2$
   (b) d block
6. (a) 3d
   n=3
   l=2
7. (a) 3f, 2d
   (b) 3rd shell has only 3 subshells – 3S, 3P, 3d
   2nd shell has only 2 subshells – 2S, 2P

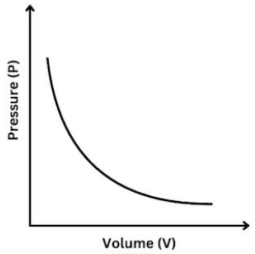
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UNIT – 4

GAS LAWS AND MOLE CONCEPT

(1 Mark Questions)

1. Write the name of the lowest temperature that can be attained by a gas?
2. How many moles in 44g CO_2
3. What is the volume of 1 mole of a gas at STP?
4. The following graph relates to which gas law?



5. What is the boiling point of water in kelvin scale?
6. Even though gaseous molecules are in constant motion, there is no energy loss due to their collision why?
7. Write the number of molecules of 1 mole of any substance?
8. Name the law which shows the relation between the number of gas molecules and its volume?
9. Write the Avogadro number

(2 Mark Questions)

1. The following related to which gas laws
    a. The size of weather balloons goes on increasing as they rise above the sea level.
    b. During summer, vehicle tyres are filled with air at lower pressure
2. Molar mass of ammonia is 17 g
    a. How many moles in 17g NH_3 ?
    b. How many molecules are in it ?
3. Filling air in foot balls
    a. At which gas law it is related?
    b. State the law?

Answers

1. Absolute zero
2. 1 mole
3. 22.4L
4. Boyle's law
5. 373 K (273+100)
6. Collisions are perfectly elastic
7. 6.022×10^{23}
8. Avogadro's law
9. 6.022×10^{23}

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UNIT - 5

ELECTRO CHEMISTRY

(1 Mark Questions)

1. Which substance is obtained at the cathode by the electrolysis of aqueous sodium chloride solution ?
2. Which metal is connected to the positive terminal of the battery during the electro plating of gold on an iron nail ?
3. What is the energy change takes place in galvanic cell ?
4. Among the following which one is an example for primary cell ?
   (Ni-cd cell, Lead – acid cell, button cell, Li-ion cell)
5. Which is the anode in Zn-Cu cell ?
6. $AgNO_3$ solutions cannot be stored in copper vessels. Why ?
7. Among the following metals, which one can displace hydrogen from Hydrochloric acid ?
   (Gold, Silver, Sodium, Copper)
8. Substance that causes oxidation is known as …….
9. A Zn rode is dipped in Copper sulphate solution. Which metal is displaced in this reaction ?
10. Which reaction takes place at anode ?

(2 mark Question)

1. A galvanic cell is constructed using Mg & Ag
   (a) Which is the cathode of this cell ?
   (b) Write the direction of electron flow.
2. Molten sodium chloride undergoes electrolysis.
   (a) Which substance is obtained at the anode ?
   (b) Write the energy change takes place in electrolytic cells ?
3. Write any two applications of reactivity series.
4. (a) What are fuel cells ?
   (c) Write any one advantages of fuel cell
5. Write any two uses of electrolysis?

(4 mark Questions)

(a) Draw the diagram of Cu-Ag cell
(b) Which is the anode ?
(c) Write the chemical equation that takes place at cathode.

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Answers

(1 Mark Questions)

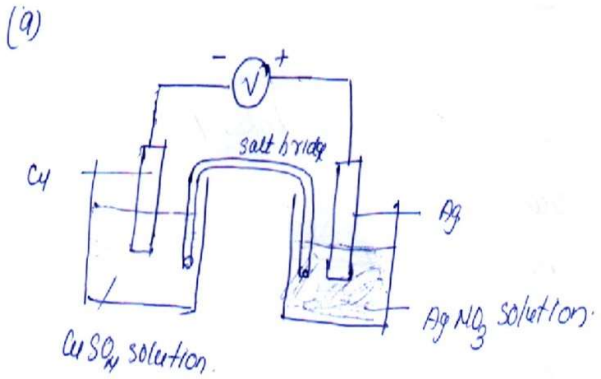
1. Hydrogen
2. Gold
3. Chemical energy changes to electrical energy
4. Button Cell
5. Zn
6. Displacement reaction takes place
7. Sodium
8. Oxidizing agent
9. Copper
10. Oxidation

(2 Mark Questions)

1. (a) Silver
   (b)From Mg to Ag
2. (a) Cl_2
   (b) Electrical energy changes to chemical energy
3. To identify which metal is being displaced, to identify oxidizing agent and reducing agent
4. (a) Fuel cells are galvanic cells designed to convert the chemical energy produced by the combustion of fuels such as hydrogen, methane and methanal into electrical energy
   (c) Electricity with high efficiency, eco-friendly
5. Production of metals, production of non-metals electroplating, pur ification of metals.

(4 Mark Questions)

1. (a)



2. (b) Cu
   (c) $2Ag^+ + 2e^- \longrightarrow 2Ag$

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UNIT - 6

METALS

(1Mark Questions)

1. Match the following

A Properties of Ore	B Method of concentration
(a) Ore has magnetic mature	(i) Leaching
(b) Ore is less denser than impurities.	(ii) Magnetic separation
(c) Ore is dissolved in suitable solution Impurities are insoluble	(iii) Froth floatation
	(iv) Levigation

Choose the correct one from the following

	a	b	c
A	(i)	(ii)	(iii)
B	(ii)	(iii)	(i)
C	(iii)	(i)	(iv)
D	(ii)	(i)	(iii)

2. Which method is used for the concentration of Copper pyrities ?
3. Name the reducing agent used in the production of Aluminium.
4. Which method is used for the extraction of metals from its oxide ore ?
5. Find the relation and fill up suitably.
   Zinc Sulphide : Roasting ; Zinc Carbonate : …….
6. Name the reducing agent used in the extraction of metals like Zn & Sn
7. Which is the Carbon steel used for the manufacturing of agricultural tools ?
8. Which are the constituent elements of brass ?
9. Name the industrial preparation of Aluminium.
10. Which is the substance used for the leaching of Bauxite.
11. At which electrode Aluminium is separated during its extraction
12. When Iron is in contact with copper, which metal undergoes corrosion ?
13. Write any two methods for the prevention of corrosion of Iron ?
14. Which alloy is used in the core of transformer ?
15. Among the following metals which one is present in nature as free state ?
    (Iron, Aluminium, Gold, Zinc)

(2 Mark Question)

1. Write any two properties of ore ?
2. Two chemical equations are given
   (i) $2\ ZnS + 3O_2 \xrightarrow{heat} 2\ ZnO + 2SO_2$
   (ii) $CaCO_3 \xrightarrow{heat} CaO + CO_2$

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(a) In second equation, which process is used for the conversion of ore into its oxide ?

(b) What is the difference between roasting and calcination ?

3. (a) Which method is used for the purification of Zn metal ?

(c) Based on which property this method is used.

4. Aluminium is industrially prepared by the electrolysis of Alumina.

(a) Why Cryolite is added to alumina during the electrolysis ?

(b) Write the chemical equation that takes place at Cathode during the electrolysis

5. (a) What are alloys ?

(c) Which alloy is used to make permanent magnets.

(3 Mark Questions)

1. Two situations are given

Situation 1 : Copper wire is wrapped over Iron nail

Situation 2 : magnesium wire is wrapped over iron nail

(a) In which situation, does the iron nail rust

(b) Write the name of the process by which the iron nail is protected here

(c) Why are Zinc or Magnesium blocks always attached to sea bridges and ship bulls ?

2. Important ores of Copper are Copper pyrites and copper glance.

(a) Which method is used for the concentration of these ores ?

(b) Which is the flux used to remove the basic impurity like iron oxide present in the ore after the concentration?

(c) What is blister copper?

(4 Mark Questions)

1. Industrial preparation of Iron is conducted at blast furnace.

(a) Which substance reduce Haematite in the blast furnace ?

(b) Write the equation for the slag formation

(c) Name the Iron obtained from the blast furnace.

(d) Which is the method used for the concentration of haematite?

Answers

(1 Mark Question)

1. B
2. Froth floatation
3. Electricity
4. Reduction
5. Calcination
6. Carbon / coke
7. Mild steel
8. Cu, Zn
9. Hall-Herault Process
10. Sodium Hydroxide Solution
11. At cathode

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12. Fe
13. Painting, electroplating
14. Silicon Steel
15. Gold

(2 mark Question)

1. Abundance, Ease of extraction, High metal content, Low production cost.
2. (a) Calcination
   (b) Calcination takes place in the absence of air and roasting in the presence of air
3. (a) Distillation
   (b) Low boiling point
4. (a) In order to decrease the melting point of alumina and to increase electrical conductivity.
   (b) $Al^{3+} + 3e^- \longrightarrow Al$
5. (a) Alloys are mixtures of two or more metals
   (b) Alnico

(3 Mark Question)

1. (a) Situation 1
   (b) Cathodic Protection
   (c) To prevent corrosion by cathodic protection
2. (a) Froth floatation
   (b) Silica
   (c) The copper lumps obtained by cooling molten copper in moulds is known as blister copper.

(4 Mark Questions)

1.(a) Carbon monoxide
(b) $CaO + SiO_2 \longrightarrow CaSiO_3$
(c) Pig Iron
(d) Levigation

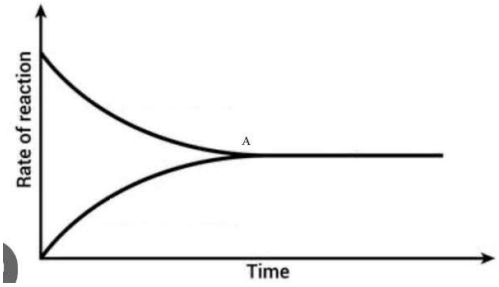
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UNIT - 7

Some Compounds of Indusrtial Importance

(1Mark Questions)

1. Which is the basic substance obtained by the heating of ammonium chloride ?
2. Find the relation and fill up suitably
   Ammonia : Haber Process ; Sulphuric Acid : ……..
3. Which are the reactants used in the laboratory preparation of ammonia?
4. A graph is given. The point A represents ……..



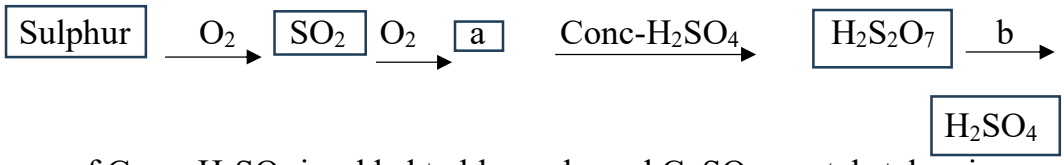
5. Write the chemical formula of oleum ?
6. Which is the catalyst used in contact process ?
7. What is the basicity of phosphoric acid (H_3PO_4)?
8. What is the name of the industrial production of Sodium Hydroxide ?
9. Among the following which is a neutral salt ?
   (NH_4Cl, Na_2CO_3, NaCl, CH_3COONa)
10. Which one is a primary nutrient among the following ?
    (Ca, K, Mg, S)

(2 Mark Question)

1. (a) During the laboratory preparation of NH_3, it is passed through quick lime why ?
   (c) Why the gas jar is kept inverted for collecting NH_3 gas.
2. (a) Write the name of the experiment used to understand the solubility of ammonia in water
   (b)Complete the chemical equation
   $NH_3 + H_2O \longrightarrow$ …….
3. Among the following which are the characteristics of chemical equilibrium ?
   (a) At chemical equilibrium the rate of forward and backward reactions are not equal.
   (b) Chemical equilibrium is static at molecular level
   (c) At equilibrium, the reactants and products are co-exist.
   (d) Chemical equilibrium is attained in a closed system.
4. What is the effect of pressure in the given chemical reaction? Write the reason.
   $H_2+I_2 \longrightarrow 2HI$

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5. Flow chart of industrial production of Sulphuric acid is given. Write a & b



6. Drops of Conc. H_2SO_4 is added to blue coloured $CuSO_4$ crystals taken in a watch glass.
   (a) Write the observation
   (b) Which property of sulphuric acid is exhibited here ?
7. (a) Which is the dehydrating agent used in the production of NH_3
   (b)Why sulphuric acid is not used as the dehydrating agent in the production of Ammonia.
8. Chemical formula of sulphuric acid is H_2SO_4
(a) What is the basicity of H_2SO_4
(b) Write the ionisation equation of H_2SO_4
9. Write the two uses of sodium Hydroxide ?
10. (a) Which acid and base react to form
    Ammonium Chloride (NH_4Cl)
    (c) The nature of NH_4Cl is ……..
    (acidic, basic, neutral)

3 Mark Question

1. $N_2+3H_2 \longrightarrow 2NH_3$ + Heat
   How the following factors affect the amount of product
   (a) Increase the concentration of Hydrogen
   (b) Decrease the pressure
   (c) Increase the temperature
2. Ammonia is essential for the growth of plants
   (a) Name the industrial preparation of Ammonia
   (b) Write the Chemical equation of this
   (c) In this reaction what is the role of catalyst after reaching the equilibrium
3. KCl is used as a chemical fertilizer
   (a) Potassium Chloride is also known as …….
   (b) Which alkali and acid react to form potassium Chloride?
   (c) Which is the nature of KCl salt?
   (Acidic, basic, neutral)

(4 Mark Question)

1. (i)A reversible reaction is given

$$2SO_2 + O_2 \rightleftharpoons 2SO_3 + \text{Heat}$$

In order to increase the rate of forward reaction what changes are to be done in the following

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(a) Pressure
(b) Temperature
(c) Concentration of Oxygen

(ii)State Le- Chatlier principle?

2. Nutrients are required for the growth of plants
   (a) Which are the nutrients required in large amount ?
   (b) Name the nutrients required in very small quantities
   (c) Write an example for a fertilizer which increase the growth of plants
   (d) Can plants utilise fertilizers that are insoluble in water ?

Answers

(1 Mark Question)

1. Ammonia
2. Contact process
3. Ammonium chloride, Calcium Hydroxide
4. Chemical equilibrium
5. $H_2S_2O_7$
6. Vanadium pentoxide (V_2O_5)
7. 3
8. Chlor- alkali process
9. NaCl
10. K

(2 Mark Question)

1. (a) To absorb moisture
   (b)Density of ammonia is less than air
2. (a) fountain experiment
   (b)NH_4OH
3. b & c
4. No effect. No of molecules of reactants and products are equal
5. (a) SO_3
   (b) H_2O
6. (a) become colourless
   (b) Dehydrating agent
7. (a) Quick lime
   (b) H_2SO_4 reacts with ammonia
8. (a) 2
   (b) $H_2SO_4 \longrightarrow 2H^+ + SO4^{2-}$
9. For the preparation of soaps and detergents,
   To produce paper, for the production of rayon
10. (a)NH_4OH & HCl
    (b)Acidic

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3 mark questions

1. (a) Increases
   (b)Decreases
   (c) Decreases
2. (a) Haber process
   (b)$N_2+3H_2 \longrightarrow 2NH_3$
   (c) No effect
3. (a) Muriate of potash
   (b) KOH & HCl
   (c)Neutral

(4 Mark Question)

1. (i) (a) Increase the pressure
   (b) Decrease temperature
   (c) Increase the concentration of oxygen

   (ii) When the concentration, pressure or temperature of a system at equilibrium is changed the system will readjust itself so as nullify the effect of change and attain a new state of equilibrium.
2. (a) N, P, K
   (b) Micro nutrients
   (c) Urea/Ammonium sulphate/Calcium nitrate
   (d) No.